Healthcare Operations Analytics

A Patient Flow & Resource Allocation Analysis

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Data Analytics Portfolio Project

Built in Snowflake — SQL Analysis

Dataset Period: April 2023 – October 2024

Executive Summary

This report analyzes 15 months of patient encounter data (April 2023 – October 2024) from a medium-sized hospital network to identify operational inefficiencies in patient flow and propose data-backed improvements. The analysis was performed end-to-end in Snowflake using SQL — covering data cleaning, exploratory analysis, and a department-and-shift-level capacity assessment.

The headline finding is that hospital-wide wait times are a systemic issue — 59.3% of patients wait longer than 30 minutes — but the underlying cause is not uniform across departments. A capacity analysis (patients per doctor, by department and shift) shows General Practice carries by far the heaviest per-doctor patient load (400+ patients per doctor), yet its wait times and satisfaction scores remain among the best in the hospital. Neurology, by contrast, has a comparatively moderate patient load but records the worst wait times and lowest satisfaction scores hospital-wide, alongside a real staffing gap (a single doctor covering the Morning shift). This distinction — volume alone does not predict patient experience, but under-resourcing in a complex-care department does — is the central insight driving this report's recommendations.

Business Problem

A medium-sized hospital network is experiencing increasing operational strain due to rising patient volumes and growing complexity in care delivery. Leadership has observed inconsistencies in how patients move through the hospital, from arrival to final disposition, but the exact causes were unclear prior to this analysis.

Hospital operations involve several interacting factors: varying patient inflow across time, department-level capacity differences, admission decision variability, and trade-offs between speed and quality of care. While patient encounter data was being captured, it had not been systematically analyzed to support decision-making.

Primary Business Question: *How can the hospital leverage patient flow data to identify operational inefficiencies, improve patient experience, and optimize resource allocation across departments?*

Dataset Summary

The dataset consists of 9,216 patient encounter records collected between April 2023 and October 2024, linked to a doctor reference table containing department and shift assignments.

Category	Fields
Patient demographics	Age, Gender, Race
Visit details	Admission Date & Time, Visit Type (Emergency / Walk-in / Scheduled)
Operational metrics	Department Referral, Wait Time
Outcome variable	Admission Flag (Admitted / Not Admitted)
Experience metric	Patient Satisfaction Score
Relational data	Doctor ID → linked to Doctor table (department, shift type)

Methodology

All work was performed in a dedicated Snowflake environment (warehouse, database, and schema created for the project). Analysis was written entirely in SQL, using CTEs for multi-step aggregation logic (favored over nested subqueries for readability), window-style aggregate functions, and multi-table joins between the patient and doctor datasets.

The analysis proceeded in three stages: (1) data quality review and cleaning, (2) exploratory SQL analysis across ten core business questions, and (3) a follow-up department-and-shift-level capacity analysis to test whether patient load was actually translating into worse patient outcomes.

Data Quality Review & Cleaning

Key data quality findings:

- Null values present in department and doctor_id fields, indicating some data collection gaps.
- Admission date was stored as a text (VARCHAR) field rather than a proper date type.
- Gender field contained at least one corrupted/concatenated entry requiring correction.
- Satisfaction score contained missing values, treated as non-response rather than an error.

Cleaning actions taken:

- Converted the admission date field to a proper DATE type.
- Replaced null department values with "Unknown" rather than dropping the rows, to preserve encounter volume.
- Corrected the identified invalid gender entry.
- Retained nulls in satisfaction score and doctor_id rather than imputing them, to avoid manufacturing data that wasn't collected.

Key Insights

1. Patient Volume Over Time

Finding: Total encounters rose from 4,338 (Apr–Dec 2023, 9 months) to 4,878 (Jan–Oct 2024, 10 months).

Interpretation: *Once normalized for the unequal period lengths, that's roughly 482 encounters/month in 2023 versus 488/month in 2024 — essentially flat. Patient demand is stable rather than sharply growing; the raw year-over-year total is misleading without this adjustment.*

2. Visit Type Distribution

Finding: Emergency: 3,167 encounters | Walk-in: 3,071 | Scheduled: 2,978.

Interpretation: *Volume is fairly balanced across the three visit types, with a slight skew toward Emergency. Because Emergency visits are the least predictable in timing, this segment deserves priority in any triage or staffing redesign.*

3. Wait Time by Department

Finding: Average wait time ranges from 33.7 minutes (Renal, lowest) to 36.8 minutes (Neurology, highest) — roughly a 3-minute spread across all departments.

Interpretation: *Wait times are consistently elevated hospital-wide rather than isolated to one department. This points to a systemic process issue (e.g. intake/triage flow) in addition to any department-specific capacity issues.*

4. Service-Level Efficiency

Finding: 59.3% of all patients wait longer than 30 minutes.

Interpretation: *This is the clearest single efficiency metric in the dataset: a majority of patients are outside a reasonable service window. Reducing this percentage should be the top operational KPI.*

5. Peak Arrival Hours

Finding: Hourly patient volume stays within a relatively narrow band (roughly 349–436 arrivals per hour) across the full 24-hour cycle.

Interpretation: *Arrivals are fairly evenly distributed rather than concentrated in sharp spikes, meaning staffing schedules should plan for sustained, near-constant demand rather than a single predictable rush.*

6. Department × Hour Bottlenecks

Finding: The highest-wait combinations are Neurology at 1am (43.6 min), Physiotherapy at 9am (42.2 min), Neurology at 11pm (41.3 min), Renal at 1pm (41.0 min), and Cardiology at 2am (40.1 min).

Interpretation: *Operational strain is concentrated in specific department-hour intersections rather than spread evenly — Neurology alone appears three times in the top bottleneck list, across both overnight and midday hours. This makes it a stronger candidate for targeted intervention than a hospital-wide policy change.*

7. Patient Satisfaction by Department

Finding: Satisfaction ranges from 3.54 (Neurology, lowest) to 3.92 (Cardiology, highest), on a 5-point scale.

Interpretation: *Neurology is the lowest-scoring department on patient satisfaction, consistent with its wait-time results above — the same department shows up as the weak point on two independent experience metrics.*

8. Admission Rates by Department

Finding: Admission rates cluster tightly between roughly 48.9% and 51.3% across departments.

Interpretation: *Admission decisions are applied with similar frequency across departments — there's no evidence any single department is systematically over- or under-admitting relative to the others.*

9. Wait Time vs. Admission Likelihood

Finding: Admission rate is 51.9% for Short Waits, 50.9% for Medium Waits, and 49.4% for Long Waits (under 15 min, 15–30 min, 30+ min respectively).

Interpretation: *The relationship is negligible — only a 2.5-percentage-point spread across the full range of wait times, trending slightly downward. Wait time is not a meaningful predictor of admission likelihood, and there's no sign that longer-waiting patients are being systematically triaged as more or less severe.*

10. Shift-Level Patient Load & Wait Time

Finding: Night shift carries the highest patient volume (3,508) and the highest average wait time (35.5 min); Morning follows (2,725 patients, 35.0 min); Evening has the lowest volume (1,783) and lowest wait (34.9 min).

Interpretation: *Night shift is both the busiest and the slowest, which warrants closer inspection — particularly since the department-level capacity analysis below shows most departments are relatively well-staffed at night on a per-doctor basis. That combination suggests the night-shift slowdown may be more of a workflow or triage issue than a pure headcount shortage.*

Department & Shift Capacity Analysis

The insights above raised a question the initial ten queries couldn't answer on their own: is high patient volume in a department actually translating into worse patient experience? To test this, doctor headcount was joined against patient volume at the department × shift level to compute a patients-per-doctor ratio, then cross-checked against each department's wait time and satisfaction results.

Highest patient-per-doctor combinations:

Department	Shift	Patients	Doctors	Patients / Doctor
General Practice	Morning	815	2	407.5
General Practice	Evening	812	2	406.0
Cardiology	Morning	299	1	299.0
Orthopedics	Evening	556	2	278.0
Cardiology	Evening	266	1	266.0

Best-covered combination (for contrast):

Department	Shift	Patients	Doctors	Patients / Doctor
Gastroenterology	Morning	256	5	51.2

General Practice carries by far the heaviest per-doctor load in the hospital — roughly 8x higher than the best-staffed combination — and this holds across both its Morning and Evening shifts, ruling out a single-shift scheduling gap as the explanation.

Interpretation: *Despite this, General Practice ranks 2nd-best on both wait time (34.8 min) and satisfaction (3.78) among all departments. The high ratio is not currently translating into a worse patient experience — likely because General Practice visits tend to be shorter and* *less complex, so volume doesn't bottleneck the same way it would in a diagnostically intensive department.*

Interpretation: *Neurology tells the opposite story. Its Morning shift is covered by a single doctor for 241 patients — a real capacity gap and a single point of failure — and Neurology is the only department that ranks worst on wait time, worst on satisfaction, and repeatedly appears among the worst department×hour bottlenecks. This is the department where a capacity shortfall is demonstrably affecting patients, not just a number that looks strained on paper.*

Business Recommendations

1. Target Neurology for additional physician coverage first.

Neurology is the only department where a real capacity gap (a single doctor covering the Morning shift) coincides with the worst wait time and lowest satisfaction score hospital-wide. This is where reallocating or adding a doctor is most likely to produce a measurable improvement in patient experience.

2. Treat General Practice's staffing ratio as a resilience risk, not an urgent outcomes problem.

General Practice's 400+ patients-per-doctor ratio is the most extreme number in the dataset, but current wait times and satisfaction there are strong. The risk is fragility: with only 2 doctors per shift, a single absence would create acute strain with no built-in redundancy. This is worth monitoring and planning for, but shouldn't be prioritized over Neurology for immediate reallocation.

3. Fix department-hour bottlenecks with targeted scheduling, not a blanket policy.

The worst wait times cluster around specific department-hour combinations — Neurology overnight and midday, Physiotherapy mid-morning, Renal early afternoon. A hospital-wide staffing increase would be inefficient; targeted shift adjustments at these specific intersections would address the bulk of the problem more directly.

4. Investigate the Night shift workflow, not just Night shift headcount.

Night shift has the highest volume and the highest wait time, yet most departments appear adequately staffed at night on a per-doctor basis. This mismatch suggests the slowdown may stem from process factors — slower lab turnaround, reduced support staff, or intake workflow — rather than a simple doctor shortage, and warrants a focused operational review before assuming more headcount is the fix.

5. Do not use wait time as a proxy for clinical urgency in resourcing decisions.

Since admission likelihood is nearly flat across wait-time buckets (49.4%–51.9%), long wait times are not a signal that patients are being appropriately deprioritized by severity — nor are they self-correcting. Reducing wait times should be pursued as a patient-experience goal in its own right, independent of admission outcomes.

Conclusion

This analysis moves the hospital's patient flow data from raw encounter records to a specific, defensible set of operational findings. The overarching pattern is that patient volume alone is not a reliable indicator of where care quality is suffering — General Practice absorbs the heaviest load in the hospital without a corresponding drop in wait time or satisfaction, while Neurology's more moderate load, combined with a genuine staffing gap, produces the worst outcomes on both measures. Prioritizing Neurology for near-term staffing attention, monitoring General Practice's resilience, and treating the Night shift as a workflow question rather than a pure headcount question together form a focused, evidence-based starting point for improving both operational efficiency and patient experience.